THE FUTURE OF

Programming

What happens to coding when AI can write the code?

Ashwin Rao · Stanford University · o9 Solutions

A talk for the curious — no coding expertise required

The question everyone is asking

"If AI can write the code…
what is left for people to do?"

The short answer:

Our job shifts from telling computers HOW … to telling them WHAT.

First — what IS programming, really?

It's writing a recipe so precise that a machine with zero common sense can follow it.

Today: the HOW

- Take 2 cups of flour.
- Add 1 egg, stir for 60 seconds.
- Bake at exactly 350° for 25 min.
- *Every. Tiny. Step.*

What we actually want

"I want a warm, fluffy chocolate cake that serves 8."

Nobody really cares about the steps —**they care about the cake.**

For 70 years, we obsessed over the HOW

Every new programming language just made the recipe a little easier to write — never the goal easier to describe.

1940s
Machine code
Talks in 1s and 0s

1970s
C
Closer to English

1995
Java
Runs anywhere

2000s
Python
Easy to work with

Today
AI Code-Gen
Simple prompts → Code

Each step made coding more accessible — but describing the actual problem stayed just as messy.

We've done this before

Remember directions before GPS?

Before

"Go 2 miles, turn left at the gas station, right after the church, merge, stay in the left lane..."

You navigated the HOW yourself.

With GPS

"Take me home."

You say WHAT you want. The machine figures out the HOW — and reroutes when things change.

AI is doing for programming exactly what GPS did for driving.

The new division of labour

HUMANS

Own the WHAT

- What problem are we solving?
- What should it actually do?
- What matters — speed, safety, cost?
- What does “done right” mean?

AI

Owns the HOW

- Writes all the actual code
- Handles the fiddly details
- Tests and checks its own work
- Does it in seconds, not weeks

So the human job gets... better

The tedious part leaves. The deeply human part stays — and grows.

Understand people

Figure out what someone truly needs — often before they can say it.

Decide what matters

Make the judgment calls: what's fair, safe, worth doing.

Bring the taste

Creativity, ethics, and good taste can't be automated.

The new superpower: asking clearly

If AI builds whatever you describe, then describing it well becomes the whole game.

"Make me an app."

Vague in, vague out.
The machine has to guess what you meant — and it will guess wrong.

"A budgeting app for students: track 3 accounts, alert when spending tops $X, private by default."

Clear in, right out.
When you're precise about WHAT, the machine nails the HOW.

Clear thinking — not typing speed — becomes the skill that pays.

A new kind of language — built for everyone

To say WHAT clearly, we won't all learn to code. New "problem languages" will let anyone describe a goal precisely.

The old way

Describing a problem rigorously meant dense mathematical notation — the domain of a few PhD specialists.

Powerful, but locked away from **almost everyone.**

The new way

Plain, structured language anyone can use — a teacher, a nurse, a student — to spell out exactly what they need.

The rigor lives under the hood.
You bring the idea; AI brings the math.

And here's the best part: it can prove it's right

When the goal is written precisely, AI can mathematically check that its solution truly matches the goal.

Today: testing = hoping

We try a bunch of examples and hope nothing breaks. Bugs slip through — we only catch the cases we thought to test.

Tomorrow: proof = knowing

The system checks the solution against the goal like a math proof — guaranteed correct for every case, not just the tested ones.

That's the leap: AI solutions you can actually trust — reliable by proof, not by luck.

How it actually works — in three steps

Think of it like working with a brilliant, lightning-fast builder.

1

You describe it

In plain words: what you want, and what really matters.

2

AI confirms the plan

It writes back a precise, checkable plan and asks about anything unclear.

3

AI builds & verifies

It produces working software and proves it matches your plan.

The journey: from a conversation to real software

Humans and AI shape the problem together. Then AI carries it all the way to a finished product.

FIGURING OUT THE PROBLEM

You + AI, together

1. **Just talk**
   Describe what you want in your own words
2. **Gather the bits**
   Notes, examples, pictures — even messy ones
3. **AI tidies it up**
   It organizes your idea and checks nothing's missing
4. **A clear goal**
   A precise description of exactly what to build

→ *hand-off*

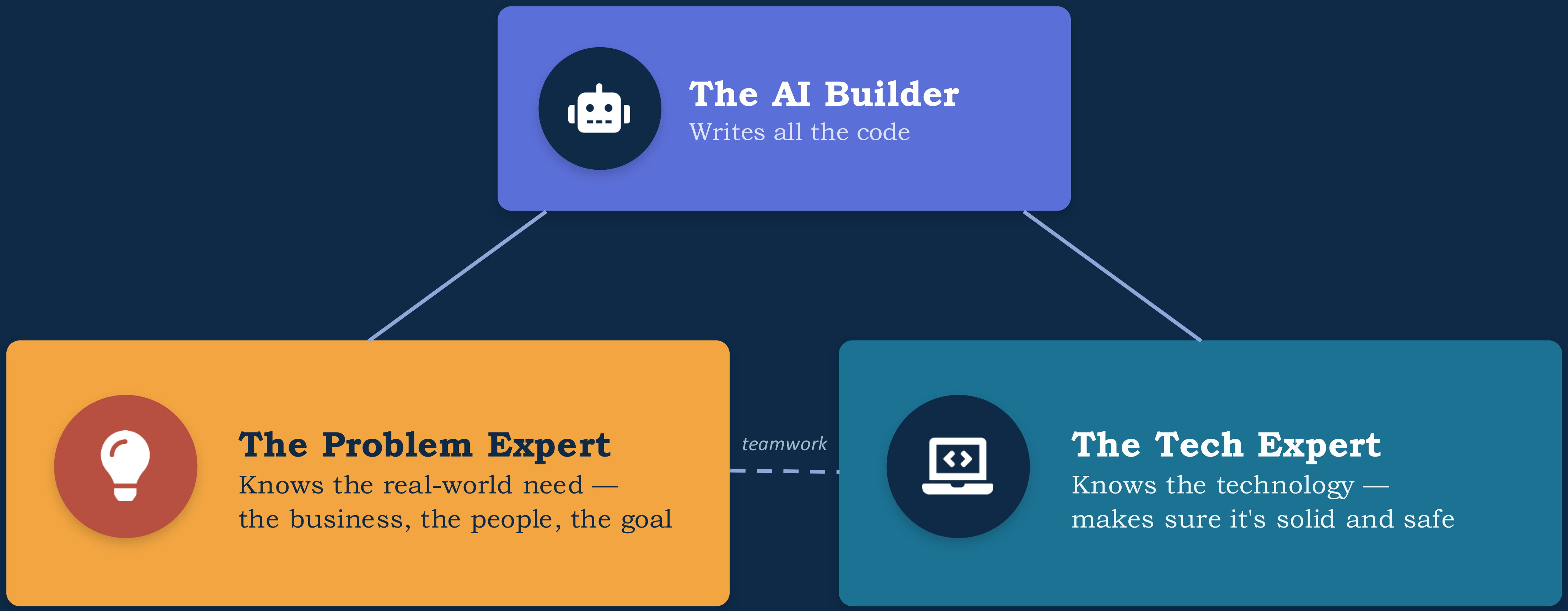
BUILDING THE SOLUTION

AI handles it end to end

1. **Design**
   AI plans how the software fits together
2. **Build**
   AI writes the actual code
3. **Check**
   AI tests it and proves it meets the goal
4. **Ship**
   AI gets it running for real people to use

Every problem now has three owners

A new way of working together. They share the goal — but only one of them writes the code.



Shared goal · shared credit · the machine does the typing.

Good news — for everyone in this room

Students

You don't need years of syntax first. Learn to think about problems — start building now.

The curious

You can turn an idea into something real without becoming a full-time coder.

Leaders

The bottleneck moves from 'can we build it?' to 'do we know what's worth building?'

A fair, honest note

This shift is real and already underway — but it's a journey, not an overnight switch.

AI isn't perfect

It still makes mistakes. Humans review and decide — judgment stays with us.

Skills evolve

The valuable skill shifts toward clear thinking and good questions.

Humans stay central

Someone has to care about the right things. That's the part machines can't own.

The one idea to take home

We're not losing the work. We're keeping the best part.

Machines get the
HOW

Humans keep the
WHAT

...and the WHAT is the part that was always worth caring about.

A great time to be curious

You don't have to become a programmer. You get to become a clearer thinker.

Practice describing what you want — precisely.

Pick something you wish existed and write down exactly what it should do.

Play with an AI tool this week.

Ask it to build something small. Notice how clearer requests get better results.

Lean into the human skills.

Curiosity, judgment, and taste are about to matter more than ever.

The future of programming isn't less human. It's more focused on what humans do best.